

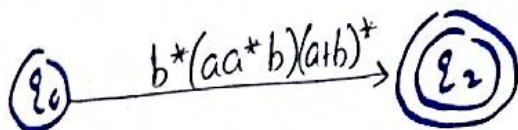
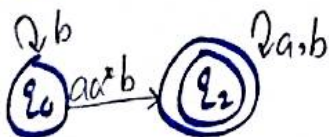
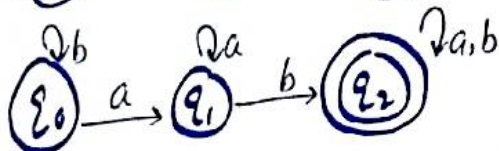
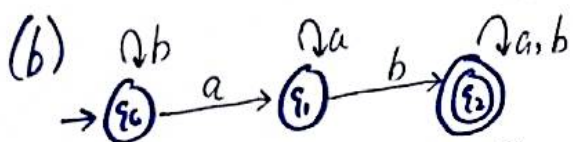
Name: Yash-Raj

Roll-no: 24k-0737.

Section: BLS-3H

Question #1 (part A):

(a) $R.E \rightarrow (a+b)^*ab$



R.E : $b^*(aa^*b)(a+b)^*$ Ans.

(v) $S \rightarrow E | HE | HTE$

$E \rightarrow 012141618$

$T \rightarrow 01112131 \dots 9$

Ans.

(vi) $S \rightarrow aA | bB$

$A \rightarrow bB | \lambda$

$B \rightarrow aA | \lambda$

Ans.

(vii) a^2n

$S \rightarrow aas | \lambda$

Ans.

(viii) $(ab)^n$

$S \rightarrow abs | \lambda$

(ix) $S \rightarrow 012 | 120 | 201$

(x) $S \rightarrow 0s1 | 1s0 | ss | \lambda$

Ans.

Question #1 (part B):

Q#1: (i) $0^* (0+1)^*$

$S \rightarrow 0A$

$A \rightarrow 1A | B$

$B \rightarrow 0B | 1B | \lambda$

Ans.

(ii) $01(0+1)^*$

$S \rightarrow 01A$

$A \rightarrow 0A | 1A | \lambda$

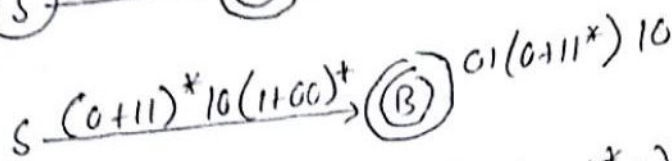
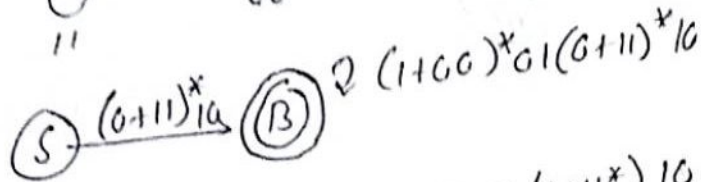
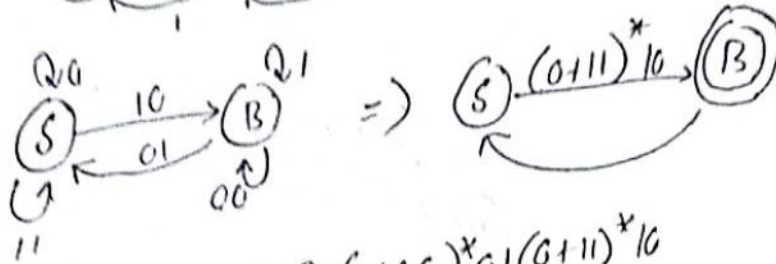
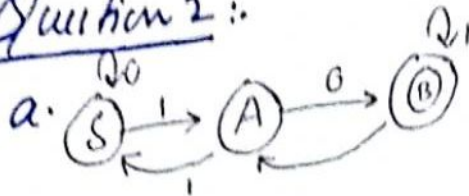
(iii) $S \rightarrow \text{letter digit}$
 $\text{letter} \rightarrow A/B/C/\dots/Z$
 $\text{digit} \rightarrow 1/2/3/\dots/9$

(iv) $S \rightarrow aAa$

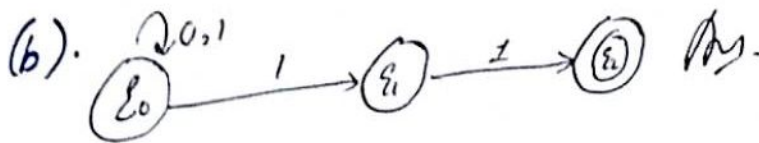
$A \rightarrow aA | bA | \lambda$

Question 1 (part A):

Question 2:



R.E = $(0+11)^*10(1+00)^*01(0+11)^*10$ Ans.



Question 1 (part B):

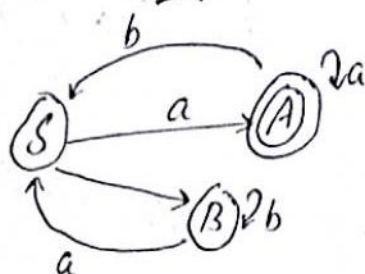
Question #2:

(a) $S \rightarrow aA / bB$

$A \rightarrow aA / bS / \epsilon$

$B \rightarrow bB / aS$

SDN.



states = $\{S, A, B\}$

initial = S

final-state = A

Ans.

Name: Yash Raj

Roll-no: 24K-0737.

Question 1 (part B):

Question #2 :-

(b) $S \rightarrow OS / 1A$

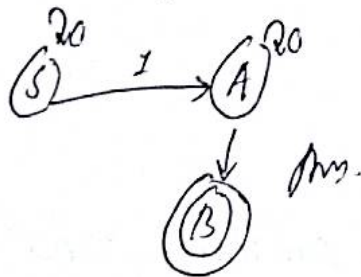
$A \rightarrow OA / 1$

sol

States: $\{S, A\}$

initial = S

final = B



(c) $S \rightarrow OA / 1B$

$A \rightarrow 1S / OA$

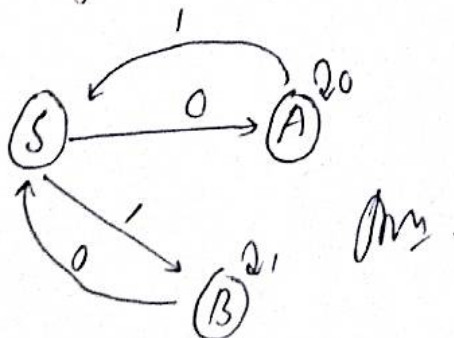
$B \rightarrow OS / 1B$

sol

States: $\{S, A, B\}$

initials S

final = ~~B~~ B.f

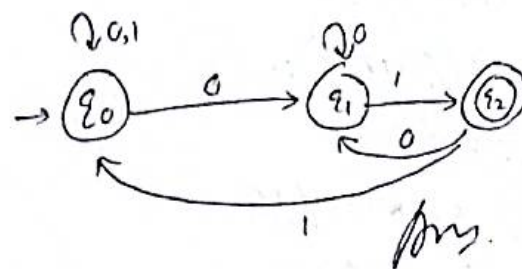


Question #3:

(a) DFA that accepts string ending with '01'

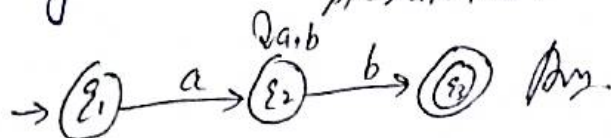
(a) Right linear grammar.

$$S \rightarrow 0S \mid 1S \mid 01$$



(b) NFA for $L = \{w \mid w \text{ starts with } a \text{ \& ends with } b\}$.

(a) Right linear $\Rightarrow$ $S \rightarrow aA$
 $A \rightarrow aA \mid bA \mid b$.

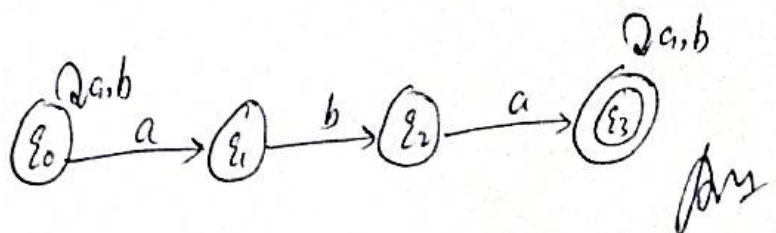


(c) NFA that all strings contain substrings 'aba'.

(a) Right linear grammar:

$$\left. \begin{array}{l} S \rightarrow aA \mid bS \\ A \rightarrow bB \mid aA \\ B \rightarrow ac \mid bs \\ C \rightarrow ac \mid bcl \end{array} \right\} \begin{array}{l} S \rightarrow as \mid bs \mid abaa \\ A \rightarrow aA \mid bA \mid \lambda \end{array}$$

(b) CFG $\therefore$ $S \rightarrow AabaA$
 $A \rightarrow aA \mid bA \mid \lambda$



Name: Yash Raj

Roll no: 24K-0737

Question #4 ::(a) Convert LLG $\rightarrow$ RLG $S \rightarrow Aa/Ab$ $A \rightarrow Ba/Bb/a$ Soln

* B is not defined here so it is not possible.

* RLG is not defined.

(b) $S \rightarrow aA/bB$ $A \rightarrow a/as/b$ $B \rightarrow b/bb/a$ Soln $S \rightarrow Aa/Bb$ $A \rightarrow a/sa/b$ $B \rightarrow b/sb/a$ Ans.Question #5::(i) $S \rightarrow asb/lab$ string = aabb(a) $P_1: S \rightarrow asb$ (i) $P_2: S \rightarrow ab \rightarrow$ put in (i) $S \xrightarrow{P_1} asb$ $asb \xrightarrow{P_2} aabb = aabb$

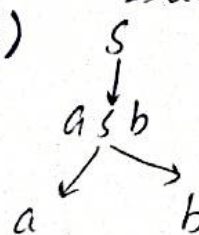
(b) left-most

1. S 2. $P_1 \Rightarrow asb$ 3. $P_2 \Rightarrow aabb = aabb$

Right most.

1. S 2. $P_1 \Rightarrow asb$ 3. $P_2 \Rightarrow aabb$

(c)



Ans.

(2) $S \rightarrow aS/ba$ string $aaabbb$.

804.

(a) Derivations.

$$P_0: S \rightarrow aS$$
$$P_1: a1 \rightarrow a1$$
$$p_2 = aac \rightarrow aash$$

$p_3: aash \rightarrow aaabb$

(b) Right most:-
1-5

1-5

$$2. \quad \underline{P_2} \rightarrow sb$$

3. $P_1 \Rightarrow a \leq b$

4. $P_1 \Rightarrow aash$

5 $P_3 \Rightarrow aaabbb$.

left-most

1-5

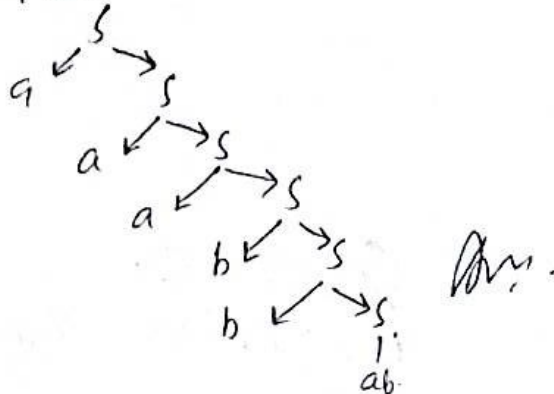
$$2 \cdot P_1 \Rightarrow 95$$

3. $P_1 \Rightarrow aas$

4 $P_2 \rightarrow aasb$

5. $\frac{P_3}{2} \rightarrow aacub$

(c) Parse Tree:-



(3) $s \rightarrow 051101$ string = 000111

502-

(a) Derivation:-

$P_0: S \rightarrow 051$

$$P_1 = 0.51 \rightarrow 0.0511.$$

$P_{25} \ 00511 \rightarrow 000111$

(b) Right master

1-5

2 - $\frac{P_0}{P_1} > 0.51$

3 - $P_1 \Rightarrow$ GOSSII.

4 - $\frac{P_2}{\rho} \rightarrow 0$ $\rightarrow 0$

left - most

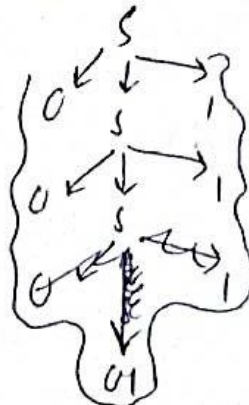
1-5

2- $\Rightarrow \text{osl}$

3 - $\frac{1}{2} \rightarrow 00811$.

4- $P_2 \rightarrow 000111$

(c) Parse-Tree



2) 000111

name: Yash Ray

Roll no: 24K-0737.

Question #6 ::

(1) $S \rightarrow (S) / \lambda$

(2)(a) $0S0 / 1S1 / \lambda$

(b) $0S0 / 1S1 / 0/1$

(3) $S \rightarrow 01A110B$

$A \rightarrow 1100A / \lambda$

$B \rightarrow 10B / \lambda$

(4) $S_1 \rightarrow 0S / 0A$
 $A \rightarrow 0A / \lambda$
 $S \rightarrow S_1 / S_2$

$S_2 \rightarrow 0S_1 / B_1$
 $B \rightarrow 2B / \lambda$

(5) $S \rightarrow AB$
 $A \rightarrow 0A1 / 01$
 $B \rightarrow 2B3 / 23$

$S_2 \rightarrow 0S_13 / C$
 $C \rightarrow 1C2 / 12$

(6) $S \rightarrow S_1 / S_2$

$S \rightarrow 0S_12 / 0S_1 / A$

$A \rightarrow 1A / \lambda$

(7) $S \rightarrow S_1 / S_2$

$S_1 \rightarrow AB / \lambda$

$A \rightarrow 0A1 / \lambda$

$B \rightarrow 2B / \lambda$

$S_2 \rightarrow CD / \lambda$

$C \rightarrow 0C / \lambda$

$D \rightarrow 1D2 / \lambda$

(8) $S \rightarrow AB / \lambda$

$A \rightarrow 0A / \lambda$

$B \rightarrow 1B22 / \lambda$

(9) $S \rightarrow 11 / S / 0S1 / \lambda$

(10) $S \rightarrow S_1 / S_2$

$S_1 \rightarrow 0S_1S_1 / 0S_1S_1 / \lambda$

Any.

Question #7 ::

Only CFG's required.

(c) $S \rightarrow (S) / \{S\} / \{S\} / \lambda$

Derivation: $(S) \Rightarrow (\{S\}) \Rightarrow (\{\{S\}\}) \Rightarrow (\{\{\lambda\}\}) \Rightarrow (\{\{\}\})$

Ans:

Part (c) :-

$$S \rightarrow a s a / b s b / a / b$$

$$A \rightarrow B / a$$

$$B \rightarrow C / b$$

(a) C is non-generating so remove

$$S \rightarrow a s a / b s b / a / b$$

$$A \rightarrow B / a$$

$$B \rightarrow b$$

→ Now A & B are not reachable so remove it.

$$S \rightarrow a s a / b s b / a / b \text{ Ans.}$$

(b) abba can't be generated by the given grammar
as $S \rightarrow a s b / b s b / a / b$.

→ Only generate odd lengths

Whereas abba is even length so abba is not part of language.

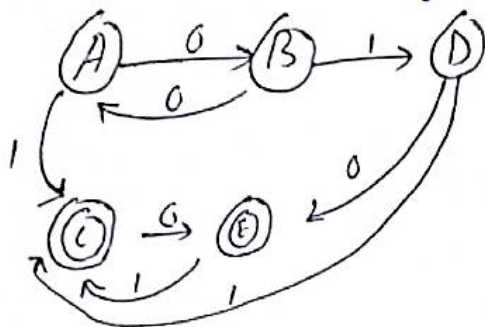
$\left. \begin{matrix} c \\ d \\ e \end{matrix} \right\}$ cancelled!

usname = Yash-Ray

Yash-Ray : 24K-0737.

Part (D) :-

Q#1 :- DFA minimization :-



partition - method :-

$P_1 = A, B, D$			
$P_2 = C, E$			
	P_1	$\left\{ \begin{array}{l} A \\ B \\ D \end{array} \right.$	$\begin{array}{l} B \ 1 \ C \ 2 \\ A \ 1 \ D \ 1 \\ E \ 2 \ C \ 2 \end{array}$
	P_2	$\left\{ \begin{array}{l} C \\ E \end{array} \right.$	$\begin{array}{l} E \ 2 \ C \ 2 \\ E \ 2 \ C \ 2 \end{array}$

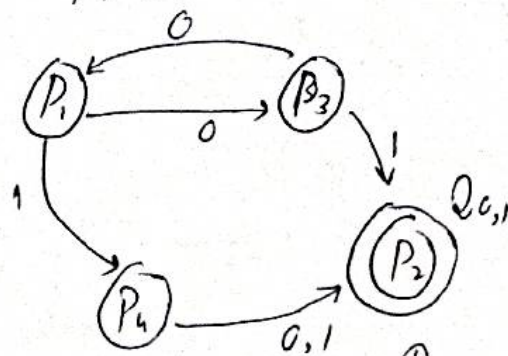
$\Rightarrow$ After

		0	1
P_1	B	A 3	D 4
	A	B 1	C 2
	D	E 2	C 2
	C	E 2	C 2
	E	E 2	C 2

$P_1 \rightarrow B$
 $P_2 \rightarrow C, E$
 $P_3 \rightarrow A$
 $P_4 \rightarrow D$

$\Rightarrow$ Minimized DFA :-

$P_1 \rightarrow B$
 $P_2 \rightarrow C, E$
 $P_3 \rightarrow A$
 $P_4 \rightarrow D$



Ans

Part (D) :

Question 2 :



Transition-table:

		a	b
q_1	q_0	$q_1(1)$	$q_0(1)$
	q_1	$q_2(1)$	$q_1(1)$
q_2	q_2	$q_3(2)$	$q_2(2)$
	q_3	$q_3(2)$	$q_3(2)$

$$P_1 \Rightarrow q_0, q_1$$

$$P_2 \Rightarrow q_2, q_3$$

		a	b
P_1	q_0	$q_1(1)$	$q_0(1)$
P_2	q_2	$q_3(2)$	$q_2(2)$
	q_3	$q_3(2)$	$q_3(2)$
P_3	q_1	$q_2(1)$	$q_1(1)$

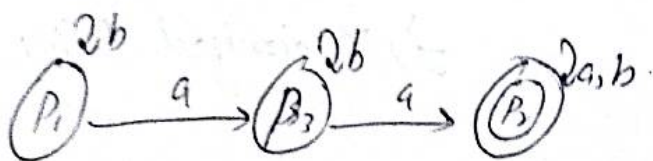
$$P_1 = q_0$$

$$P_2 = q_2, q_3$$

$$P_3 = q_1$$

minimized DFA :-

$$P_1 \Rightarrow q_0 \quad P_2 \Rightarrow q_2, q_3 \quad P_3 = q_1$$



Transition-table:-

	a	b
P ₁	P ₃	P ₁
P ₂	P ₂	P ₂
P ₃	P ₃	P ₃

Am